

INTERLOCK VOLATILITY CREDENTIAL EXTRACTION

Behavioral candidate for Volatility3 plugins that extract SAM hashes or cached domain credentials from a Windows memory image.

PRODUCTION CANDIDATE

CONFIDENCE: MEDIUM

SOURCE-CONFIRMED BEHAVIOR

retain facts, isolate interpretation

DeviceProcessEvents

direct telemetry boundary

HUNT · VALIDATE · TUNE

no automatic malicious verdict

REPORT DATE

07 August 2026

VERSION

v1.0

PRIMARY TELEMETRY

Microsoft Defender XDR ·
DeviceProcessEvents

ATT&CK

T1003.002 · T1003.005

Evidence-led dossier. Query status remains an untested
implementation sketch until executed in the target environment.

INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Detect the credential-extraction plugin, not the tool filename.

Sophos EIR directly observed Interlock executing Volatility3 with `windows.hashdump.Hashdump` and `windows.cachedump.Cachedump`. These canonical plugin strings describe credential extraction and remain visible even if the Volatility launcher is renamed.

OPERATIONAL VALUE

High-specificity credential-access visibility.

The logic avoids hashes, domains and arbitrary thresholds. It creates a direct endpoint alerting candidate before stolen hashes are used for privilege escalation or lateral movement, while preserving an explicit DFIR exception path.

9.0

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Candidate

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- Sophos observed both plugin commands on Patient Zero during a March 2026 Interlock intrusion.
- Hashdump was used to extract NTLM and LM hashes plus local user information.
- Cachedump was used to extract cached domain usernames and hashes.

ANALYTICAL ASSESSMENT

What the evidence supports

- The plugin strings express credential-extraction intent more durably than `vol.exe`.
- Either plugin invocation is sufficient for an actionable credential-access event.
- Authorized DFIR, assessment and malware-analysis use is the principal benign overlap.

DETECTION ASSUMPTIONS

What must be validated

- `ProcessCommandLine` is complete and retained.
- Approved Volatility activity can be identified by actor, asset and case record.
- The plugin identifiers are not removed or materially changed.

Confidence boundary

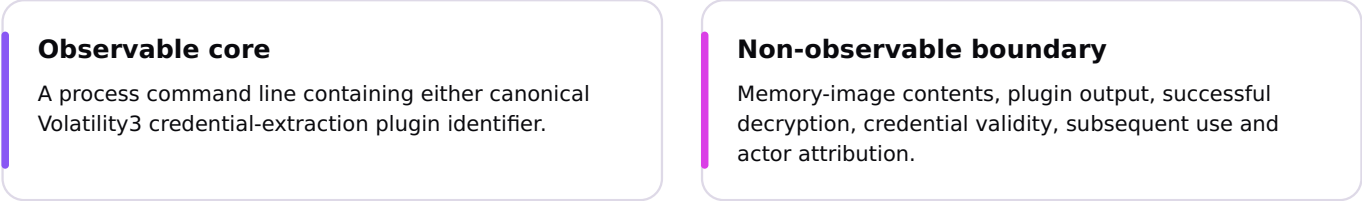
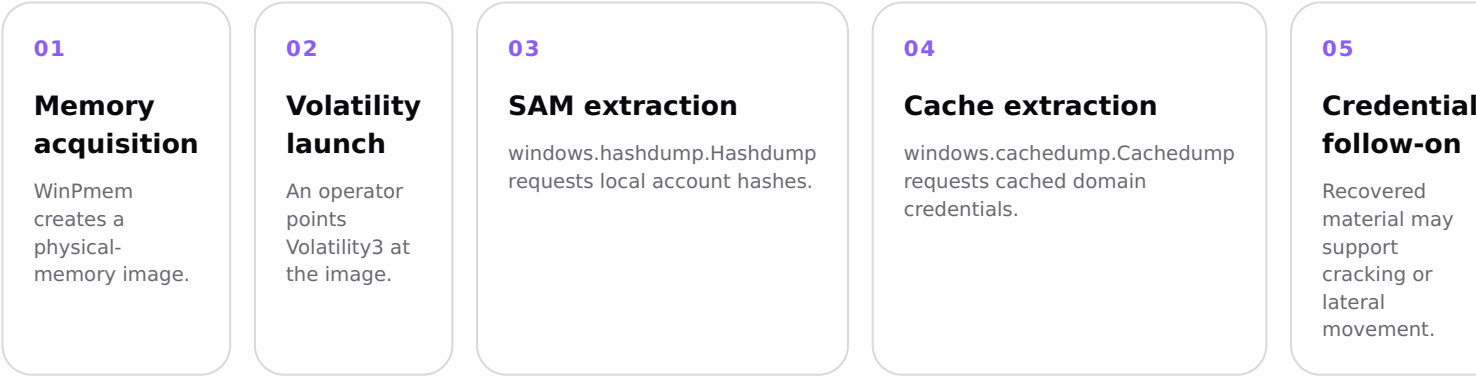
A match proves the plugin was requested, not that extraction succeeded or that Interlock is responsible. Production promotion requires controlled replay and a documented DFIR allowlist.



THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

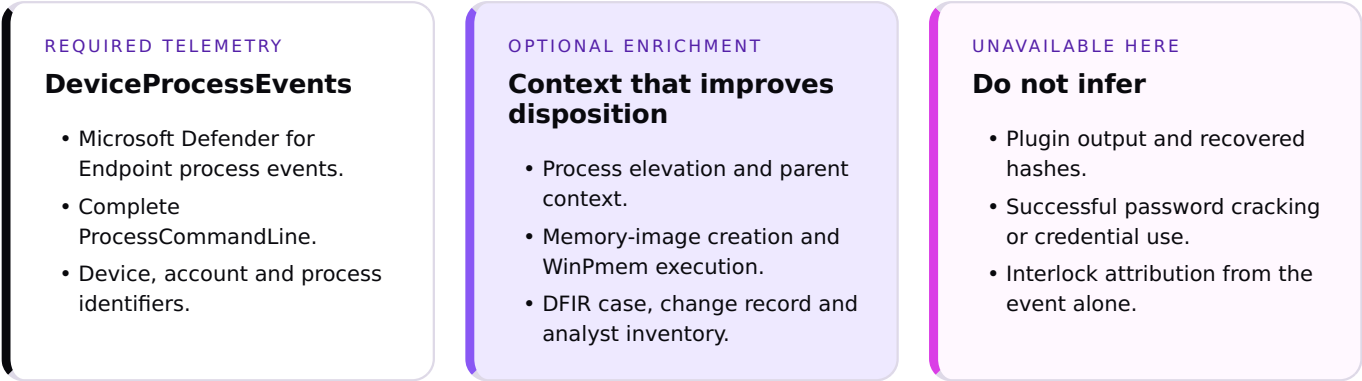
ATTACK FLOW



DETECTION HYPOTHESIS

Credential extraction should expose the selected Volatility plugin.

If an attacker uses Volatility3 to extract SAM hashes or cached domain credentials, DeviceProcessEvents should show windows.hashdump.Hashdump or windows.cachedump.Cachedump in ProcessCommandLine. Legitimate activity can be distinguished through approved DFIR assets, analysts, case records and acquisition provenance.





TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR · DeviceProcessEvents

Contains process creation events with command line, executable, account, elevation, parent, hashes, device and unique process identifiers.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

The table can establish plugin invocation. It cannot establish successful extraction, inspect the memory image or reveal the resulting credentials.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Process event	DeviceProcessEvents	Primary execution evidence.	REQUIRED
Command	ProcessCommandLine	Both evidence-bearing plugin identifiers.	REQUIRED
Device	DeviceId, DeviceName	Endpoint pivot and asset policy.	REQUIRED
Account	AccountDomain, AccountName	Authorization and DFIR ownership check.	REQUIRED
Process key	ProcessUniqueId	Stable investigative correlation.	REQUIRED
Elevation	ProcessIntegrityLevel, ProcessTokenElevation	Privilege context.	OPTIONAL
Parent	InitiatingProcessFileName, InitiatingProcessCommandLi ne	Launcher and operator context.	OPTIONAL
Event identity	Timestamp, ReportId	Raw-event pivot.	REQUIRED

Data quality requirements

- Confirm full command-line collection.
- Retain raw events on workstations and forensic hosts.
- Inventory approved DFIR tools, analysts and case windows.
- Verify identifiers through a controlled Volatility3 replay.

Known failure conditions

- Command lines are missing or truncated.
- A wrapper hides the plugin argument.
- The attacker uses another credential-dumping method.
- Authorized DFIR activity cannot be reliably identified.

DETECTION OBJECTIVE

Find direct Volatility credential-extraction intent.

Select process events containing either source-observed canonical plugin identifier, independent of the launcher filename.

ANALYTICAL LOGIC

Direct filters and investigable output.

Two direct substring predicates; no IOC, filename dependency, join, sequence, threshold, aggregation, rarity or automatic attribution.

KQL · UNTESTED CANDIDATE

// Interlock - Volatility credential-extraction plugins
// Classification: Production candidate - Untested implementation sketch
// Confidence: Medium
// Primary platform: Microsoft Defender XDR
// Telemetry: Microsoft Defender for Endpoint DeviceProcessEvents
//
// Sophos EIR directly observed both plugin invocations in an Interlock intrusion.
// The plugin strings describe SAM hash extraction and cached domain credential
// extraction. Do not suppress approved DFIR activity by filename alone: validate
// the actor, device, case/change record, memory image provenance and execution time.

DeviceProcessEvents
| where ProcessCommandLine contains "windows.hashdump.Hashdump"
 or ProcessCommandLine contains "windows.cachedump.Cachedump"
| project Timestamp,
 DeviceId,
 DeviceName,
 AccountDomain,
 AccountName,
 FileName,
 FolderPath,
 ProcessCommandLine,
 ProcessIntegrityLevel,
 ProcessTokenElevation,
 ProcessUniqueId,
 InitiatingProcessFileName,
 InitiatingProcessFolderPath,
 InitiatingProcessCommandLine,
 InitiatingProcessUniqueId,
 SHA1,
 ReportId
| order by Timestamp desc

REVIEW FIELDS · BASELINE LOCALLY · VALIDATE BEFORE PRODUCTION

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested candidate	None	None	DeviceId + ProcessUniqueId	Manual triage

What a match means

A process requested a Volatility3 plugin designed to extract local or cached domain credential material and requires immediate authorization review.

What a match does not prove

Successful extraction, valid credentials, malicious operator, follow-on access, ransomware deployment or Interlock attribution.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the raw event.
Verify command line, executable, account, device, time, process key and elevation.
- 02

Establish authorization.
Match the actor and asset to an active DFIR, assessment or malware-analysis case.
- 03

Validate image provenance.
Identify the memory image, acquisition tool, source host and evidence handling.
- 04

Scope related execution.
Search for WinPmem, alternate Volatility plugins, credential tools and suspicious parents.
- 05

Assess credential exposure.
If unauthorized, isolate the host and rotate potentially exposed local and domain credentials.

Escalation gate
Escalate immediately when no approved case explains the invocation, especially on ordinary workstations or servers and when memory acquisition, suspicious logons or lateral movement corroborate it.

Primary benign overlap
Incident responders, penetration testers and malware analysts legitimately use the same plugins. Tune only with verified analyst identity, dedicated assets and bounded case windows; never suppress by vol.exe or tool hash globally.

INVESTIGATION PIVOTS

Host
Role, coverage and neighboring alerts.

Actor
Account ownership, logon session and privilege.

Launcher
Path, hash, parent and provenance.

Memory image
Creation time, source and acquisition tool.

Credentials
Accounts resident on the captured host.

Follow-on
Authentication, lateral movement and privilege changes.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Incident response	An analyst may extract hashes to investigate compromise.	Require active case, approved analyst, target asset and time window.
Security assessment	A tester may validate credential exposure.	Require authorization, scope and change record.
Malware or forensic lab	Training and research may invoke both plugins.	Allowlist isolated lab assets with ownership controls.

Baseline plan

- Search 30-90 days.
- Inspect every match and raw command line.
- Map approved analysts, labs and forensic jump hosts.
- Measure events that lack reliable ownership context.

Validation procedure

- Replay both plugins against a benign test image.
- Confirm ProcessCommandLine preserves canonical identifiers.
- Test a renamed launcher and a non-credential plugin.
- Review historical matches with DFIR owners before enabling alerts.

TEST CASES

CASE	EXPECTED RESULT
Canonical Hashdump invocation	Expected match with complete process context.
Canonical Cachedump invocation	Expected match with complete process context.
Volatility benign process listing plugin	No match.
Renamed launcher with canonical plugin	Expected match because filename is not required.

Hunting Initial query	Candidate Current status	Production Measured precision	Optimized Stable feedback
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Success

Both controlled credential plugins match, non-credential plugins do not, and every legitimate match can be explained with durable authorization attributes.

Rejection

Command lines omit plugin identifiers or authorized use cannot be separated without broad, fragile exclusions.

Failure

DeviceProcessEvents coverage or process command-line telemetry is unavailable.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1003.002 / T1003.005	OS Credential Dumping: SAM / Cached Domain Credentials	Directly covers invocation of the Volatility3 plugins used to extract SAM hashes and cached domain credential material.

Detection coverage summary

Strong for these canonical Volatility3 plugins even with a renamed launcher; weak for wrappers that hide arguments or alternative credential-dumping tools.

Residual risk

Successful output and later credential use require file, identity and authentication telemetry, while actor attribution remains outside the event.

ANALYST CONCLUSION

A durable credential-access candidate with a narrow benign exception.

Direct incident-response evidence, official plugin documentation and documented Defender fields support a Production candidate. It remains an untested implementation sketch until replay confirms command-line visibility and the organization proves that legitimate DFIR use can be bounded without broad suppression.

REFERENCES

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DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v1.0	07 August 2026	Candidate	Detection Engineering

Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.